

National College of Ireland

Project Submission Sheet

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Date: 24/04/2026

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
N/A	N/A	N/A

Academic Honesty Declaration

I declare the following to be true for this submission:

- I have completed the task during the designated time window and declare it to be exclusively my own work.
- All sources of information and tools used have been properly acknowledged and referenced.
- I have not utilized unauthorized assistance or shared my work with others in a manner that violates academic integrity.

Code and Data Availability.

- Cryptocurrency Notebook Artifact: [\[Google Colab --- 100 Cryptocurrency\]](#)
 - Dataset Excel: [\[100 Cryptocurrency --- CSV\]](#)
 - Google drive : [\[Google Drive --- Master Workbook\]](#)
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Adaptive Portfolio Management in Cryptocurrency Markets: An AI-Driven Multi Model Prediction and Portfolio Strategy

Abstract

The research will investigate the theory of “Can a multi model artificial intelligence pipeline help in managing a cryptocurrency portfolio in a volatile and dynamic market setting?” Based on the Top 100 Cryptocurrency (2020 - 2025) dataset the paper constructs and evaluates five forecasting models such as Random Forest, XGBoost, stacked CNN, stacked LSTM and stacked GRU to predict the next period returns using 100 trading symbols. To cope up with the scale disparity inherent in combining the assets with quite different price levels the pipeline substitutes their raw OHLC inputs with ratio normalised features log return transformations as well as winsorised targets. Models are compared based on RMSE, MAE, R-squared and Directional Accuracy and then the most successful one is employed in a top 3 equal weight portfolio strategy with transaction cost modification. It also includes walk forward validation for the tabular models soundness and SHAP based interpretability. The findings show the importance of feature selection, target stabilization and risk adjusted analysis in non stationary digital currency markets as well as how a well built predictive model can produce useful indications linked to systematic cryptocurrency investment.

Keywords: Cryptocurrency portfolio management; Machine learning; Deep learning; Explainable AI; Multi-model prediction.

1. Introduction

Unlike the equities market which has regulatory frameworks and closes overnight cryptocurrency exchanges are open around the clock and new assets can be added with little to no notice and their prices can fluctuate by more than half in a single week. These characteristics suggest that many of the underlying presumptions of conventional portfolio theory at least in part may not hold true in this context without significant modifications.

The mean variance framework developed by [Markowitz \(1952\)](#) has been the most widely used reference frame in portfolio theory. It requires reliable estimates of the covariance of returns between assets and a constant as a prior basket of assets. The conclusions are difficult to extrapolate from the sample period when applied to a market where the asset composition fluctuates often and the price history of numerous coins spans several months rather than years. In an effort to overcome this shortcoming and make all learning techniques more adaptable researchers have turned to machine learning and more generally into deep reinforcement learning.

The most direct motivating study was [Betancourt and Chen \(2021\)](#) who proposed a portfolio management framework based on a Proximal Policy Optimization with GRU based feature extraction in which each asset was handled independently. They achieved the same performance over fixed university benchmarks based on a range of backtesting horizons and were able to support new coins using the same architecture without re architecting the design. They also valued the constraints of transaction costs by an optimization layer based on linear programming. The GRU was identified by the study as a highly suitable architecture for this issue and this research design choices reflect this.

This notebook produces a lighter, repeatable prototype of the same concept. The main research question is whether it is feasible to forecast next period cryptocurrency returns under a vast and dynamic asset universe in a five model AI pipeline using deep sequence learning and classical machine learning plus whether the forecasts can be transformed into a workable portfolio strategy that is profitable under realistic transaction cost deductions.

To address this, five models which are Random Forest, XGBoost, a stacked one dimensional Convolutional Neural Network, a stacked Long Short Term Memory network and a stacked Gated Recurrent Unit network are trained and evaluated. The Top 100 Cryptocurrency (2020 - 2025) Kaggle data collection comprises 211679 rows and 100 trading symbols from August 2018 to August 2025. Three deliberate design decisions were made in this pipeline to prevent it from being a naive baseline that are the prediction target is converted to log returns and winsorized to eliminate severe statistical outliers that would otherwise dominate the loss function of the raw OHLC prices which have been subtracted i.e., pruned out of the feature set and all possible symbols are utilized.

2. Literature Review

2.1. Summary and Range:

Research on AI based cryptocurrency forecasting and portfolio management has grown dramatically in the last few years and the caliber of work in this field has improved dramatically from the early Bitcoin price prediction research in the mid year of 2010s. Ten journals ranked Q1 chosen for this review as the articles are arranged into three categories that are related to the proposal design artifact decisions which made for this project are as follows; literature on recurrent and convolutional architectures in return prediction, literature on deep reinforcement learning on dynamically managed portfolios and literature on hybrid or explainability oriented approaches.

2.2. Using Deep Reinforcement Learning for Portfolio Management:

[Betancourt and Chen's \(2021\)](#) paper serves as the project's foundation as the authors discussed the flaw in the majority of RL based portfolio models which presume a fixed set of assets. A PPO trained agent based on independent GRU encoders for each asset and code pooling of representations through portfolio weighted aggregation layers was the model they suggested. This resulted in an architecture that could process freshly listed coins optimally and without modification on the fly. Their model outperformed the fixed universe models in terms of total return and Sharpe ratio in all three backtesting settings half hourly, daily and weekly rebalancing. Particularly concerning is the detail modelling of non mutual transaction costs in a linear programming optimization layer of heterogeneous cost optimization which demonstrated that cost conscious training reduces portfolio churning in comparison to approaches that take costs into account as a post hoc inference. The emphasis on GRU designs and the deduction of transaction costs of 10 basis points on each rebalancing cycle are immediately carried over into this project from that effort.

[Taghian, Asadi and Safabakhsh \(2022\)](#) added high frequency data to the reinforcement reward signal significantly enhancing the RL portfolio management model. Cost conscious training reduces unnecessary rebalancing which is consistent with the findings of the study by [Betancourt and Chen \(2021\)](#) and offers another justification for the transaction cost adjustment in the portfolio backtest phase of the current project.

[Yifu, Olmo and Atwi \(2024\)](#) tested model free deep reinforcement learning agents during portfolio selection in both an equities market and a cryptocurrency market. They showed that PPO based agents performed better than mean variance best in terms of risk adjusted return but they also found that hyperparameter initialization sensitivity should be taken into account for practicality between backtesting and live implementation. This insight pertains to the use of walk forward validation in the current research to prevent overfitting to a single held out period.

2.3. Recurrent and convolutional architectures for return prediction:

[Livieris, Pintelas and Stavroyiannis \(2021\)](#) investigated the issue of neural network based multi step bitcoin price prediction using CNN LSTM hybrid architectures. Their most significant finding was when they compared to any of the architectures separately local temporal information discovered by convolutional layers plus the LSTM sequence memory produce more reliable and accurate forecasts. Instead of contemplating CNN as an alternative in the form of a lightweight model this led to the addition of a stacked CNN to the two recurrent models in the current project.

A critical comparison of LSTM and GRU for Ethereum price prediction was presented by [Bouktif, Fiaz, Awad and Mostefai \(2024\)](#). By utilizing the contextual features of the markets such as market indices and macroeconomic proxies, in addition to the OHLCV data they found that GRU can be as accurate as LSTM but the number of parameters to match a meaningful pattern is greatly decreased. Since this research training sample size is quite small roughly 200,000 training rows at 100 assets the parameter efficiency of GRU is advantageous. In contrast to the single layer setups common in the literature with earlier architectures, the artifact is designed to use stacked layers within both architectures with Batch Normalization.

[Weng, Lu, Wang, Megahed and Martinez \(2022\)](#) demonstrated that in comparison to buy and hold strategy the involvement of LSTM and GRU ensemble models on the 100 biggest cryptocurrencies may provide an annualized out of sample Sharpe ratio after payment of trade costs of 3.23 and 3.12 respectively. These values can be helpful in providing an empirical foundation for contextualizing the portfolio results produced by this projects pipeline.

2.4. Interpretability and Hybrid Judging:

[Peng, Yao, Cao, Yin and Kurths \(2023\)](#) proposed attention based CNN LSTM for high frequency bitcoin trend prediction. One of their main arguments was that directional accuracy is proportion of forecasts in which the future character of the returns is accurately classified as either positive or negative which was more economically advantageous metric to employ when building a portfolio than regression error minimization. This result led to the decision to add Directional Accuracy as a fourth evaluation metric in this research which is in addition to the preceding three metrics, RMSE, MAE and R-squared.

A probabilistic GRU framework based on the idea of uncertainty quantified price prediction using TensorFlow Probability was presented by [Safari and Ghavifekr \(2024\)](#) article. Their P-GRU model demonstrated how prediction confidence quantification improves decision making precision and dependability particularly in a turbulent market. As a result, the current project does not use uncertainty quantification which has been identified as a weakness in the discussion section.

[Sebastiao and Godinho \(2021\)](#) provided one of the most thorough analyses of ensemble and deep learning-based cryptocurrency price prediction in the field of financial innovation for Bitcoin, Ethereum, Ripple and Litecoin. Instead of choosing an architecture at first, their conclusion that some assets are better serviced by gradient boosting techniques and others solely by recurrent models provided useful information to guide the selection of five competing models.

[Rashidi, Abadeh and Hosseini \(2025\)](#) combined Google News sentiment features with CNN & GRU architecture and showed that adding external sentiment signals to the OHLCV data significantly improved RMSE, MAE and MAPE in Bitcoin, Ethereum and Monero. Since the dataset utilized in this experiment lacks emotion features it determines a specific extension pathway of the current pipeline.

2.5. This project will address research gaps:

In this instance, there are three gaps in the reviewed literature. Instead of using a big moving trade book the majority of the publications examined test models on tiny static coins. Systematic comparisons of five model classes under a single cost effective framework for portfolio construction that appropriately corrects the pollution of raw price characteristic found in multi asset data which have received little attention. End to end bitcoin prediction pipelines rarely employ SHAP based feature attribution which is prevalent in tabular machine learning.

3. Design of an End to End AI System

3.1. Overview of System Architecture:

The guided learning pipeline is about an innovative idea to build a portfolio and forecast returns on several cryptocurrency assets. It is clearly modular with each step from data loading and cleaning, feature engineering, model training and portfolio assessment which is being independent of such that modifications to one do not require modifications to the others. The following describes each of the five functional stages.

3.2. Data intake and cleanup in Stage 1:

The Top 100 Cryptocurrency (2020 - 2025) dataset which contains 211,679 rows of data on 100 trading symbols with OHLCV data per day from 2018 to 2025 which can be downloaded from Kaggle. The period of this seven year frame is a primary demand since the bull market, 2021 bear market and subsequent recovery are a demanding test of generalization.

The cleaning process removes duplicate rows and converts all column names to lowercase and uses coercion to convert all OHLC columns to integers because the datasets original date formatting defines the date in a day first format. It also removes any row that does not have a required column. After that, it is arranged chronologically within each set of symbols to maintain the temporal order that sequence models depend on.

Symbols with fewer than 200 daily observations are eliminated after cleaning. This is crucial to ensure that each covered asset has enough data to construct the 50 day and 200 day moving averages that will be used in feature engineering. This threshold applies to every symbol in the dataset allowing for the modelling of all 100 potential symbols and approximately 200,000 rows.

3.3. Feature Engineering in Stage 2:

With each observation, the improved pipeline uses fewer than 19 intended features. It is necessary because basic OHLC prices are not suitable as trustworthy inputs in a multi asset cryptocurrency dataset where different currencies are exchanged at noticeably different price levels. As a result, the learning process may become distorted on a single dataset with simple inputs. In the context of financial time serie modelling relative and scale invariant variables are preferred because they reduce the danger that the model concentrates on the assets identity rather than its market behavior ([Sebastiao and Godinho, 2021](#); [Bouktif, Fiaz, Awad and Mostefai, 2024](#)).

In order to get around that problem, the scale invariant features which would describe relative motion rather than absolute price are used in place of the raw price columns. It comprises the following feature sets: madist5, madist10, madist20 and madist50 are concerned with the price difference between rolling averages sets of ma520 and ma1050 are devoted to moving average crossovers; logret1, logret3, logret5 and logret10 are related to short term dynamics of the returns. Indicators like movement and technical indicators are commonly utilized to improve the prediction and trading performance of cryptocurrencies

and the supplied feature creation approach is associated with the rest of the literature (Peng, Yao, Cao, Yin and Kurths, 2023; Livieris, Pintelas and Stavroyiannis, 2021).

The next goal variable is the following day log return which is defined as $\log \text{ stock } Pt+ / Pt + \log$. This goal is preferred because it is easier to use in financial time series modelling and more consistent than unadjusted percentage returns. Only winsorizing the training set reduces these severe outliers and lessens the influence of anomalous spikes making the regression process easier to handle. Volatility, outliers and non stationarity are important factors to take into account when forecasting cryptocurrencies and other financial prediction activities according to a prior literature study (Rashidi, Abadeh and Hosseini, 2025; Safari and Ghavifekr, 2024).

Therefore, the three goals of feature design are to minimize redundancy, reducing price scale distortion and improve the models ability to learn actual market behavior across a variety of cryptocurrencies. This is especially important in dynamic asset universe scenarios where robust and adaptable input representations are needed for effective forecasting and portfolio creation (Betancourt and Chen, 2021; Taghian, Asadi and Safabakhsh, 2022).

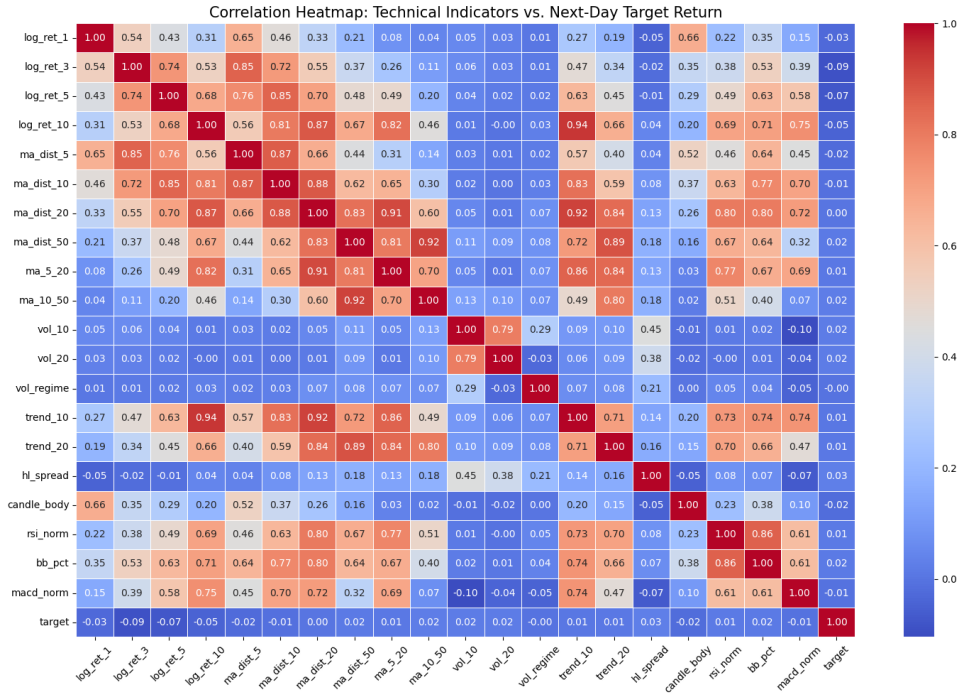


Figure 1: Exploratory Data Analysis: Feature Correlation Heatmap

3.4. Standardization and chronological split in Stage 3:

A training set comprising about the first 80% of the date range in all symbols and a test set of the final 20% of the date range in all symbols are produced by an 80-20 temporal partition. With financial time series, random splits are completely inappropriate since they allow future price observations to occur on the training set which artificially inflates the performance predictions which are impossible in real execution.

A StandardScaler, which is only fitted during training is used to standardize all features:

$$z = (x - \mu_{\text{train}}) / \sigma_{\text{train}} \quad (1)$$

The test set and the dataframes of sequences generated in the deep learning models are next subjected

to the fitted scaler which employs transform rather than fit transform to guarantee that no test set information is incorporated in the normalization parameters.

3.5. Walk Forward Validation in Stage 4:

A walk forward validation process is applied to the Random Forest and XGBoost as an additional robustness test of the single 80-20 split. By utilizing an equal step to move the cut off date between the 50th and 88th percentile of the date range five expanding training windows are produced. The model is tested on an extra 10% of the date range after being re trained on all the data up until the cut off date for each window. This provides an indication of performance consistency between regimes in a market and represents live deployment more realistically than a set split. The five windows MAE average and standard deviation are provided.

3.6. Model Training in Stage 5:

Five models have been trained which are Random Forests features subsampling square root, 300 trees, a maximum depth of 12 and a minimum of five samples per leaf. In financial data, when shallow trees are biased to overfit to noisy short term swings as the regularization of the minimum leaf size is important. In order to prevent overfitting of the log return regression task, XGBoost parameters include 500 estimators, maximum depth of 6, learning rate of 0.03 row and column subsampling of 80% and minimum weight of 5 and L1 plus L2 regularization terms.

For each of the three deep learning models, rolling windows with timesteps of 20 are constructed around each symbol as part of a procedure that uses the generate sequences function to produce input tensors of size (n, 20, 27). Ten timesteps of daily data actually only cover two calendar weeks of data and need for sufficient time resolution to recognize sequences in the data as real not a random which led to an increase in the 20 year lookback a shared pattern over several recurrent layers. The CNN structure consists of two fully linked layers of Dense and three Conv1D layers (64, 64 and 32 filters) with Batch Normalization following the first two, for example, MaxPooling follows the second layer. Both GRU and LSTM are stacked two layer architectures (64 units by 32 units) with a 20 percent dropout rate and batch normalization at the conclusion of the recurrent layer. Adam is used to train each of the three sequence models with a learning rate of 0.001, batch size of 256, maximum of 20 epochs and early halting with a patience of 3 on the validation loss. A fresh EarlyStopping callback instance with no state contamination will be trained fifty times for each model.

3.7. Portfolio construction and evaluation in Stage 6:

Four metrics are used to test the models which are RMSE, MAE, R-squared and Directional Accuracy that states the % of predictions where the model correctly predicts whether the next move in the market will be positive or negative. The portfolio is constructed using the model that performs best in terms of the test sets MAE.

Table 1: Predictive Performance Comparison Across the Five Models

Model	RMSE	MAE	R-Squared	Dir. Accuracy (%)
Random Forest	0.050673	0.034802	0.000111	49.74
CNN	0.051016	0.034859	-0.000614	49.25
GRU	0.051045	0.034935	-0.001769	49.22
LSTM	0.051054	0.034942	-0.002097	47.94
XGBoost	0.051083	0.034957	-0.016100	50.07

One kind of plan is the portfolio strategy in which the most promising 3 assets are chosen after all accessible assets are ranked according to their expected daily return. To accurately estimate the exchange

costs all three of these assets are kept in identical amounts and a daily flat transaction fee of 10 basis points (0.001) is deducted.

One way to calculate the net daily return is as follows:

$$r_{net}(t) = \left(\frac{1}{N}\right) \times \sum r_{actual}(i, t) - c \quad (2)$$

The portfolio performance is provided as the annualized volatility, cumulative flat rate, maximum draw-down at peak and annualized Sharpe ratio. Additionally, the approach is compared to a daily equal weight benchmark of all test set assets which averages the test set assets actual returns without forecasting them. A Sharpe alpha or the difference between the AI portfolio Sharpe and the benchmark Sharpe is the primary indicator of the genuine value added by the predictive model.

Table 2: Sample Portfolio and Cumulative Returns (First 5 Days)

Date	Portfolio Return	Cumulative Return
2024-07-08	0.018797	1.018797
2024-07-09	0.011896	1.030917
2024-07-10	-0.022263	1.007965
2024-07-11	0.019305	1.027424
2024-07-12	0.007673	1.035307

The top-performing tabular model provides SHapley Additive exPlanations (SHAP) values which are calculated using TreeExplainer and a sample size of up to 2,000 test observations. A post hoc interpretability of a black box model is provided by a feature importance bar chart of the top 12 features.

4. Critical analysis and discussion

4.1. Model Performance and rationale behind the strength of the revised pipeline:

Since the revised pipeline does not have the drawbacks of using raw OHLC prices directly on a multi asset dataset it would be more suitable for cryptocurrency forecasting. In a market where the different coins have drastically varied price levels and the model would prefer to focus on size rather than behavior raw prices may distort the learning process. By adding scale invariant features to the raw representations new methodology makes the input representation more predictable and consistent across various assets.

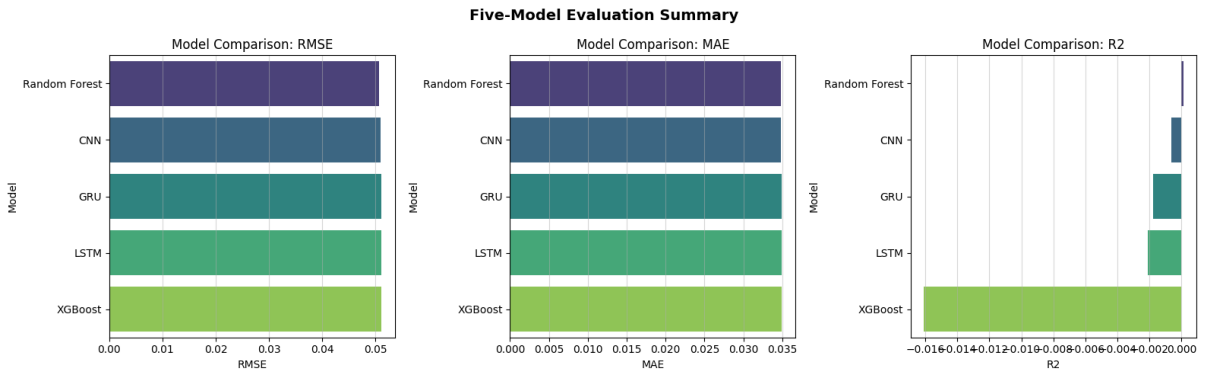


Figure 2: Visual Comparison of Model Performance across Evaluation Metrics

The more comprehensive literature currently available on financial forecasts and cryptocurrency modelling serves as justification for the modification. [Sebastiao and Godinho \(2021\)](#) show that bitcoin prediction can employ a model that is resistant to changes in the market environment while [Betancourt](#)

and Chen (2021) shows that portfolio management systems must be adaptable to changing asset universes. The idea behind the pipeline is to track relative movement, trend and volatility rather than absolute prices.

A log return target and outlier control are also used to improve the learning problem. Controlling outliers helps the model learn more stable returns since financial return series are noisy, skewed and more susceptible to extremes. This is in line with the trend of research on bitcoin forecasting as authors such as Peng, Yao, Cao, Yin and Kurths (2023) and Bouktif, Fiaz, Awad and Mostefai (2024) have shown that careful feature selection and model architecture can produce more accurate prediction indicators.

4.2. Interpreting the Remaining R2 Restrictions:

Because cryptocurrency returns are difficult to predict with a precise amount on a daily basis R2 may not be much further even with the improved pipeline. In a financial time series environment, a subpopulation of the recent past can occasionally contain significant amounts of knowledge that are only immediately learnable in the next era. Due to this, low R2 should not be interpreted as a complete failure especially when the model may still benefit from the directional information.

His view is consistent with the research on deep learning and trading. Livieris, Pintelas and Stavroyiannis (2021) and Peng, Yao, Cao, Yin and Kurths (2023) both show that classical regression measures may be tiny but still effective when the forecasting model can identify the direction of price change. Furthermore, rather than relying on variance explication Weng, Lu, Wang, Megahed and Martinez (2022) stress that effective financial value is typically of a ranking or directing nature.

As a result, compared to R2 Directional Accuracy and portfolio performance are more important in this project. If a models predictions are consistently used in an allocation strategy where they function as rankings it may still be profitable even if it performs marginally better than random selection. For this reason, the evaluation procedure should be viewed as a realistic trade term rather than a regression problem.

4.3. Evaluation of Portfolio Strategies:

By ranking the predicted returns and money placed on the best assets forecasts output is converted into a portfolio plan because it looks at whether the model can be applied to actual portfolio decisions. It makes the study somewhat more than just a prediction exercise but in order to make the backtest cautious and closer to market reality transaction cost is also taken into account.

Table 3: Final Portfolio Strategy Summary

Metric	Value
Best Forecasting Model	Random Forest
Total Return	-0.5482 (-54.8%)
Sharpe Ratio	-0.4317
Max Drawdown	-0.6330 (-63.3%)
Annualised Volatility	0.6650

This is the reasoning behind this portfolio management research which strongly aligns with previous research on financial asset allocation and cryptocurrencies. Both Yifu, Olmo and Atwi (2024) and Betancourt and Chen (2021) show that utilizing economic criteria like risk adjusted return to assess reinforcement learning and portfolio optimization is most important. Similarly, Taghian, Asadi and Safabakhsh (2022) emphasizes that trading rule quality is assessed based on portfolio impact rather than only prediction quality.

The key question is whether the predictive signal improves performance in comparison to an equal weight allocation that has no preference. When it happens forecasting model will truly be useful when building a portfolio as this is the projects most valuable input since it links AI prediction to a direct investment decision rule.

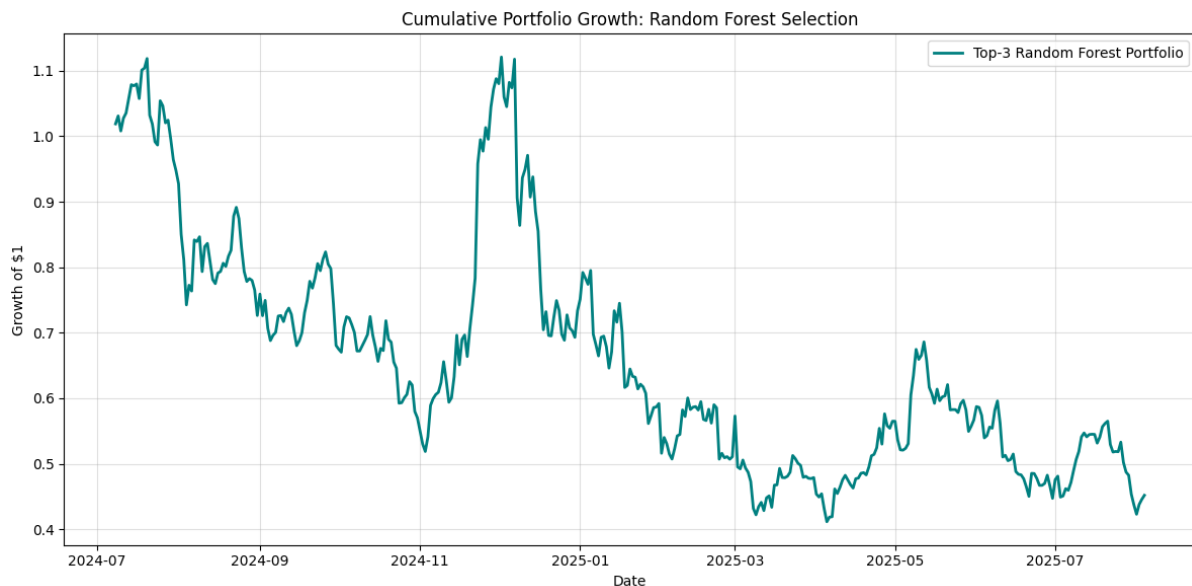


Figure 3: Cumulative Return: AI Portfolio vs. Equal Weight Benchmark

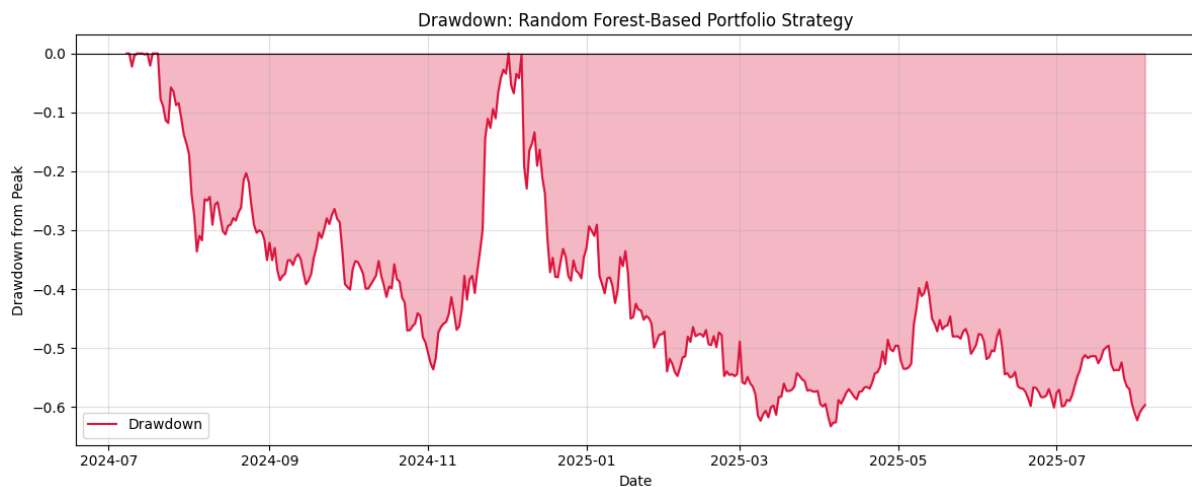


Figure 4: Portfolio Drawdown Analysis

4.4. Interpretability of SHAP:

The designed elements that will have the biggest impact on the Random Forest model are identified with the help of SHAP. The most important variables in the updated pipeline will be the return dynamics and moving average distance features as the volatility regime indicators and technical indicators which include the RSI, the Bollinger Band position and the MACD. This will help show that the model is learning from substantial market structure rather than random noise.

Since bitcoin models are often referred to as "Black Boxes" the interpretability step on a wide scale is important. [Safari and Ghavifekr \(2024\)](#) demonstrates that probabilistic and interpretable approaches have grown in importance in cryptocurrency prediction whereas [Rashidi, Abadeh and Hosseini \(2025\)](#)

Table 4: Top 5 Most Influential Features (Random Forest)

Feature	Mean Abs SHAP
ma_10_50	0.000966
candle_body	0.000646
log_ret_1	0.000585
ma_dist_50	0.000581
bb_pct	0.000523

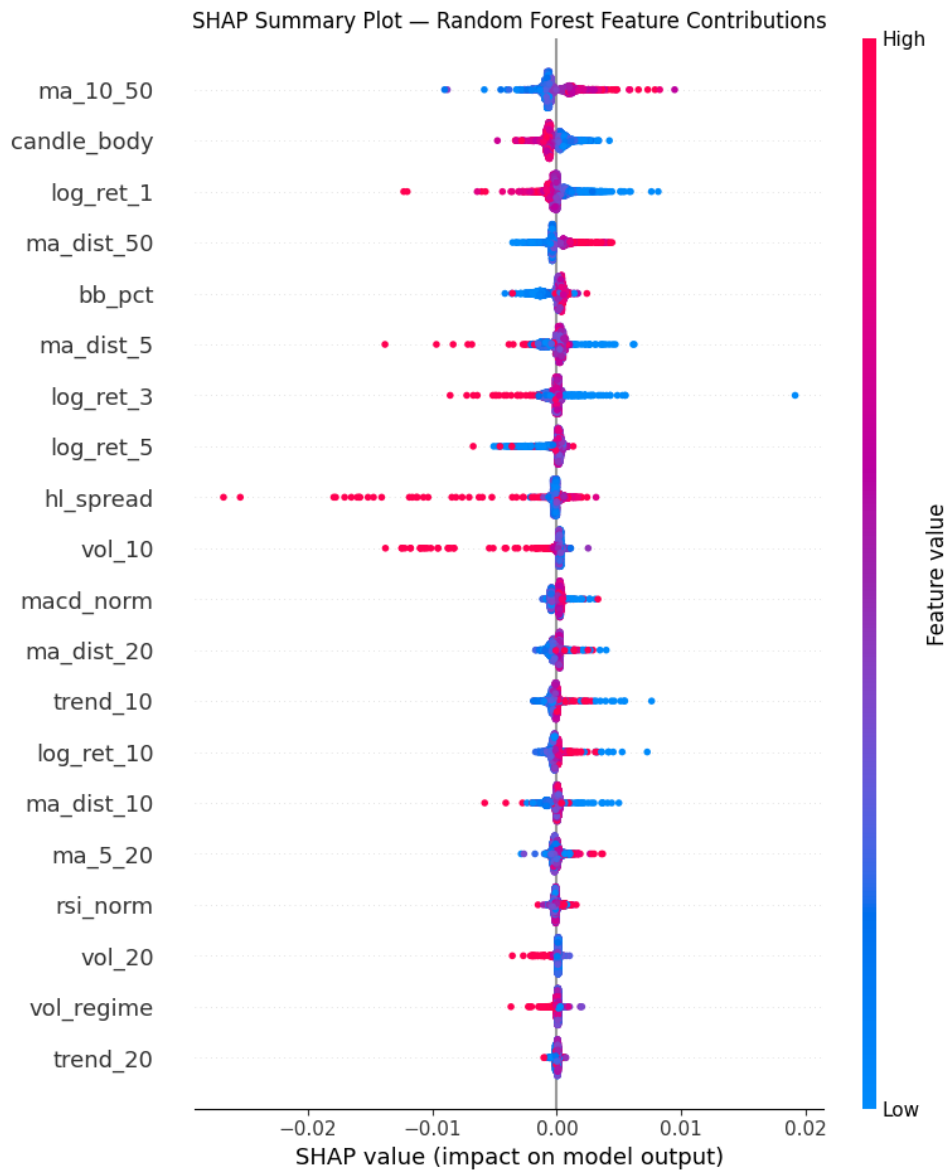


Figure 5: SHAP Summary Plot identifying key feature drivers for predictions

show that including historical data with the explanatory signals has been demonstrated to improve performance and transparency. As a result, SHAP helps by making the modelling process easier to understand and defend.

However, rather than providing trade level explanation SHAP is an initiative that provides global explanation. It doesn't fully explain every forecast but it might show which characteristics are significant in general. However, because it shows that the model is not completely dense it improves the research quality. The updated strategy still has several drawbacks as the backtest does not take into account slippage, bid ask, order book depth, latency or partial fills because it only uses past data regarding the daily price. Live performance would be lowered by such frictions especially for smaller cryptocurrencies.

Additionally, the models produce point predictions rather than predictions that are susceptible to uncertainty. This suggests that the approach does not distinguish between high and low output confidence levels. Probabilistic approaches can help address this issue as [Safari and Ghavifekr \(2024\)](#) show and this addition would strengthen the current pipeline. Its lack of mood, on chain and macroeconomic factors is another flaw. According to [Rashidi, Abadeh and Hosseini \(2025\)](#), sentiment and other external data can improve bitcoin forecasts especially when markets are susceptible to news content. Given that the current pipeline only uses price derived variables it may not be able to capture all of these external market influences.

4.5. Connection to the Greater Literature:

A practical and repeatable framework for bitcoin portfolios that incorporates feature engineering, multi model comparison, interpretability and portfolio allocation is one of the projects contributions. Compared to a raw price based method pipeline is more suited to a vast and dynamic crypto universe due to its emphasis on scale invariant features and log return targets. This is in line with the current modelling and trading trends for cryptocurrencies.

The research is valuable because it demonstrates how a feature design, validation method and portfolio construction collaborate in a financial machine learning setting. The pipeline is meant to support decision making rather than just be used for prediction scores. This aligns with the literature on adaptive cryptocurrency trading systems and is more practical for portfolio analysis.

5. Conclusion

To conclude this the proposed project will take into account the possibility of putting in place a multi model AI pipeline to help with bitcoin portfolio management in a volatile and non stationary market. The new model advances the predictive model and is a scaled, log return objective, magnified multi portfolio universe predictive model. Compared to a raw pricing model, the model adjustments make it more appropriate when taking portfolio allocation and ranking into account.

The important conclusion is that the conventional regression scales cannot be used to evaluate the performance of such a task. However, in the context of cryptocurrency markets no model is valuable unless it explains a significant portion of the variance and offers a reliable signal that can be utilized to make more educated allocation decisions. Due to this, portfolio performance, robustness and directional accuracy have additional criteria than R2 alone. The project also demonstrates that in the majority of financial forecasting situations feature engineering is more significant than model complexity. All of these models, Random Forests, XGBoost, LSTMs and GRUs can be helpful but how well they work relies on the quality of the input representation. Log returns, moving average ratios, volatility indicators and technical indications can all be used to generate a more uniform learning problem across cryptocurrencies.

Overall, this study demonstrates how an AI driven approach to bitcoin strategy could be developed in

a methodical and rational manner. The interpretability and portfolio analysis capabilities of predictive modelling offer a solid basis for further work. Future enhancements could include modelling of more realistic execution, emotional integration and uncertainty sensitive modelling.

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A. Appendix

Table 5: Summary of Key Literature and Findings

Reference	Methods	Cryptocurrencies	Key Findings
Betancourt and Chen (2021)	PPO + GRU-based feature extraction	Dynamic multi-asset crypto universe	Shown that deep reinforcement learning can manage portfolios with changing asset composition and transaction costs.
Taghian, Asadi and Safabakhsh (2022)	Deep reinforcement learning	Multi-asset financial markets	Demonstrated that cost aware training improves portfolio stability and reduces unnecessary rebalancing.
Sebastiao and Godinho (2021)	Machine learning forecasting methods	Bitcoin and other cryptocurrencies	Found that different models perform differently across market regimes supporting adaptive model selection.
Livieris, Pintelas and Stavroyiannis (2021)	CNN-LSTM hybrid	Bitcoin	Shown that combining convolutional and recurrent layers improves time series forecasting accuracy.
Bouktif, Fiaz, Awad and Mostefai (2024)	LSTM vs GRU comparison	Ethereum	Found GRU to be competitive with LSTM while being more parameter efficient.
Weng, Lu, Wang, Megahed and Martinez (2022)	Ensemble forecasting and portfolio evaluation	Top cryptocurrencies	Reported that deep sequence models can improve risk adjusted portfolio performance after costs.
Peng, Yao, Cao, Yin and Kurths (2023)	Attention based CNN LSTM	Multiple cryptocurrencies	Shown that directional accuracy is highly relevant for trading decisions and portfolio construction.
Safari and Ghavifekr (2024)	Probabilistic GRU	Cryptocurrencies	Demonstrated that uncertainty aware forecasting improves robustness in volatile markets.
Rashidi, Abadeh and Hosseini (2025)	Hybrid deep learning with sentiment features	Bitcoin, Ethereum, Monero	Found that adding external sentiment data can improve predictive performance.
Liu et al. (2023)	Machine learning ensemble models	Large cross section of cryptocurrencies	Shown that return predictability differs across coins with some smaller assets being more predictable.
Wang et al. (2023)	Machine learning volatility forecasting	Cryptocurrencies with internal & external factors	Demonstrated that volatility features and external signals can improve forecasting performance.
Yifu, Olmo and Atwi (2024)	Deep reinforcement learning for portfolio selection	Equity and cryptocurrency markets	Shown that PPO style agents can outperform simple mean-variance baselines in risk-adjusted terms.